

CSE 299

RideRent Pro

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Abstract—The RideRent Pro system is a smart vehicle rental platform that allows users to rent vehicles conveniently through a mobile application. The system provides online and manual booking facilities for different types of vehicles such as bikes, private cars, and minibuses. Users can view vehicle details, availability status, pricing, fuel type, ratings, and driver information before making a reservation. The platform integrates secure online payment methods including mobile banking and card-based transactions. RideRent Pro aims to improve accessibility, reduce rental complexity, and provide a seamless transportation experience for customers.

Index Terms—Key Words : Vehicle Rental System, Mobile Application, Online Booking, Smart Transportation, fleet Management, Driver Management, QR code Authentication, Customer Relationship Management, Vehicle availability Tracking, Real time Booking, Location Based Services, Cloud Based System, Payment Gateway, RideRent Pro

I. INTRODUCTION

Transportation plays a significant role in modern society. Many people require temporary access to vehicles for personal, business, or travel purposes. Traditional vehicle rental services often involve lengthy paperwork, manual processes, and limited accessibility.

RideRent Pro is designed to simplify the vehicle rental process through a digital platform. Customers can browse available vehicles, compare prices, select drivers, and complete bookings through a mobile application. The system enhances user convenience while improving operational efficiency for rental service providers.

II. PROBLEM STATEMENT

The existing vehicle rental process is often manual, time-consuming, and inefficient. Customers frequently face difficulties in checking vehicle availability, comparing rental prices, selecting drivers, and completing secure payments. Many rental services lack a centralized digital platform that provides real-time information and convenient booking facilities. As a result, customers experience delays and inconvenience while service providers struggle with efficient management of vehicles and bookings. Therefore, there is a need for a smart and integrated vehicle rental system that can automate and simplify the entire rental process.

III. OBJECTIVE

The main objectives of the RideRent Pro system are:

- * To develop a smart vehicle rental platform.
- * To provide real-time vehicle availability information.
- * To support both online and manual booking facilities.
- * To integrate

- secure online payment methods.
- * To provide driver selection options for customers.
- * To improve customer convenience and satisfaction.
- * To simplify vehicle rental management for administrators.
- * To ensure efficient branch-based vehicle operations.

IV. LITERATURE REVIEW

The vehicle rental industry has experienced significant growth due to advancements in digital technologies and mobile applications. Existing platforms such as Uber Rentals, Hertz, and Avis provide vehicle booking services through online systems. These platforms offer convenience and accessibility but often focus on limited functionalities. Many systems do not provide integrated driver selection, branch-based management, QR-based vehicle access, or combined manual and online booking facilities.

Several studies have highlighted the importance of digital transformation in transportation services. Researchers have emphasized the benefits of automated booking systems, secure payment gateways, and real-time vehicle tracking in improving customer satisfaction and operational efficiency. RideRent Pro aims to incorporate these features into a unified platform to provide a more comprehensive rental solution.

V. PROPOSED SYSTEM

RideRent Pro is a mobile-based vehicle rental management system designed to provide a seamless rental experience for customers and administrators. The system allows users to register, log in, browse available vehicles, compare vehicle details, make reservations, select drivers, and complete payments through secure digital methods.

The proposed system includes customer management, vehicle management, booking management, payment processing, branch management, and driver management modules. Customers can choose vehicles based on category, price range, fuel type, ratings, and availability. The system also supports QR-based verification and smart access mechanisms to improve security and efficiency.

VI. METHODOLOGY

The development of RideRent Pro follows a structured methodology. First, users register and create an account using their personal information. After successful authentication, customers can browse available vehicles and view detailed information regarding pricing, fuel type, brand, and ratings.

The user selects a vehicle and verifies its availability status. If required, the customer may also select a driver from the

available driver list. The booking is then confirmed through either online payment or manual payment methods. Upon successful payment, a confirmation message is generated. The vehicle can be collected from a designated branch and returned to any available branch within the specified rental period. The system automatically calculates additional charges for late returns.

VII. SYSTEM FEATURES

The RideRent Pro system provides the following features:

- User Registration and Login
- Vehicle Search and Selection
- Real-Time Availability Checking
- Vehicle Details and Rating Display
- Driver Selection Facility
- Online and Manual Booking Options
- QR-Based Smart Card Verification
- Secure Digital Payment System
- Booking Confirmation Notifications
- Customer Reviews and Ratings
- Branch Management System
- Late Return Charge Management

VIII. EXPECTED OUTCOMES

The implementation of RideRent Pro is expected to improve the efficiency of vehicle rental services by reducing manual operations and enhancing customer convenience. The system will provide real-time access to vehicle information, simplify booking procedures, and ensure secure payment processing. It is also expected to increase customer satisfaction, improve resource utilization, and support the digital transformation of transportation services.

IX. FUTURE WORK

Future enhancements of RideRent Pro may include GPS-based vehicle tracking, AI-powered vehicle recommendations, dynamic pricing models, automated maintenance scheduling, real-time traffic integration, and predictive analytics for demand forecasting. The system may also be extended to support electric vehicle rentals and smart city transportation infrastructures.

X. CONCLUSION

RideRent Pro is an intelligent and technology-driven vehicle rental management system designed to modernize traditional transportation services through automation, digitalization, and smart decision-making. The system integrates advanced features such as real-time vehicle availability monitoring, secure online payment processing, QR-based vehicle access, branch management, driver selection, customer review analysis, and AI-powered recommendation mechanisms within a unified platform.

Unlike conventional rental services, RideRent Pro aims to provide a seamless and user-centric experience by enabling customers to make informed rental decisions based on vehicle ratings, pricing, availability, and personalized suggestions. The

integration of artificial intelligence can further enhance the platform by recommending suitable vehicles, predicting customer preferences, optimizing fleet utilization, and improving operational efficiency.

This project demonstrates how emerging technologies can be utilized to transform the vehicle rental industry into a smarter, more efficient, and customer-oriented ecosystem. In the future, RideRent Pro has the potential to evolve into a comprehensive smart mobility solution through the incorporation of GPS tracking, predictive analytics, IoT-enabled vehicle monitoring, and advanced AI-driven services.